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Product Requirements Document (PRD)

GroNatural Analysis Software - Post-Alpha Feedback

Version: 1.0
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Based on: Client survey responses (12 respondents, Jan 28 – Feb 14, 2026)

Table of Contents

Product Requirements Document (PRD) *1*
GroNatural Analysis Software - Post-Alpha Feedback 1
1. Executive Summary 2
2. Feedback Summary by Theme 2
2.1 What Users Valued Most 2
2.2 What Users Want Changed or Removed 2
2.3 What Users Want Added (Ranked by Frequency) 3
3. Requirements 3
P0 - Critical (Must ship next) 3
P1 - High Priority (Next iteration) 5
P2 - Medium Priority (Subsequent iterations) 7
P3 - Low Priority / Future Consideration 8
4. Data Accuracy Improvements 9
5. UI / UX Improvements 9
6. External Data Sources to Integrate 10
7. Success Metrics 10
8. Out of Scope (This Phase) 11
9. Appendix: Raw Survey Scores 11

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1. Executive Summary

The alpha of the GroNatural agricultural data visualization platform was demoed to the GroNatural team. Twelve stakeholders completed a post-demo survey. This PRD distills their feedback into actionable requirements for the next phase of development.

Overall sentiment: The platform's core concept - visualizing crop acreage on an interactive map - was well received. UI satisfaction averaged **8.6 / 10** and data robustness scored **7.8 / 10**. However, data accuracy confidence was lower at **6.1 / 10**. This metric was driven primarily by the "usable land" density metric, which every respondent flagged as misleading or broken. Resolving this, adding organic acreage data, and displaying actual planted acres are the highest-leverage changes.

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2. Feedback Summary by Theme

2.1 What Users Valued Most

- Visualizing crop/value distribution on a map rather than in tables
- Sorting by crop, then finding the land owner and farm data
- Ease of use: "similar to other map-based programs"
- Ability to modify queries in real-time (filters, layer toggling)
- Seeing acres by crop, by state, by county in one place

2.2 What Users Want Changed or Removed

- **"Usable land" density metric**: nearly universal complaint. Rangeland, forestry, and non-arable land inflate denominator, producing misleading density values. Must be reworked or removed.
- **Soybean classification**: soybeans are currently grouped with specialty crops. Should be reclassified.
- **Unclear variable names**: some labels/options are not intuitive to the team.

2.3 What Users Want Added (Ranked by Frequency)

1. Organic vs. conventional acres (7 respondents)
2. Actual planted acres: not satellite-estimated potential (5 respondents)
3. Historical crop rotation data: 3 years (4 respondents)
4. Retailer / distributor / ag-retail location pins (4 respondents)
5. Better crop groupings (row crop, annual, tree fruit, berry, etc.) (3 respondents)
6. Multi-county / area selection with aggregated summary (2 respondents)
7. CRM integration (2 respondents)
8. Rate-of-change / growth trend metrics (2 respondents)
9. Soil type overlay (1 respondent)
10. Ag chemical label lookup / Agrian integration (1 respondent)

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3. Requirements

P0 - Critical (Must ship next)

3.1 Rework the Density / "Usable Land" Metric

Problem: The current density calculation divides crop acreage by satellite-derived "usable land," which includes rangeland, forestry, and other non-cropland. In the PNW and mountain states this produces near-zero densities that do not match reality.

Requirements:

- Replace the denominator with **total cropland acreage** (sum of all CDL agricultural pixel categories in the county), not total land or satellite-estimated "usable" area. This will instead give the share of specialty/selected crops out of all crops in the selected region.
- Rename the metric from "Crop Density" to something unambiguous, e.g., **"Share of Cropland"** or **"% of Crop Acres."**
- Provide a tooltip or info icon explaining what the metric represents.
- If a secondary "arable land potential" metric is retained for leadership use, it must exclude: roads, buildings, wetlands, federal parks, land-management areas, and residentially zoned land. Label it clearly as a separate, exploratory metric.

Data source: CDL raster (already in pipeline). Summing all agricultural CDL codes per county produces the correct denominator.

3.2 Show Actual Planted Acres (Absolute Values)

Problem: Users want to see absolute planted acreage - not proportions or satellite-estimated potential. Several respondents reported confusion about whether numbers represented real plantings or assumptions.

Requirements:

- Default the primary data display to **absolute acres** (or hectares where requested) per crop, per county.
- Label clearly as "Planted Acres (CDL 2024)" so users understand the data age.
    - Provide clarity on CDL accuracy / show average error from census.
- Proportional / density views remain available as a secondary toggle, not the default.

3.3 Add Organic Acres

Problem: Organic acreage is a top priority for 7 of 12 respondents. The platform already geocodes organic farms (for California) but does not yet distinguish organic vs. conventional breakdowns in the main UI.

Requirements:

- Integrate the **USDA Organic Integrity Database** to determine certified organic acres by crop, by county.
- Display an "Organic / Conventional" toggle or filter in the sidebar.
- When viewing a county or parcel, show organic acres alongside conventional acres.
- Where exact organic acreage per crop is unavailable, display organic farm count as a fallback and label accordingly.

Data source: USDA Organic Integrity Database (already partially integrated). Extend the pipeline to aggregate acreage by crop and county.

3.4 Reclassify Soybeans

Problem: Soybeans are currently categorized as a specialty crop, skewing data for a user focused on actual specialty crops.

Requirements:

- Move soybeans out of the specialty crops grouping.
- Create a separate classification (e.g., "Row Crops" or "Field Crops") or exclude soybeans from the default view with an option to include them.
- Confirm final list with GroNatural.

P1 - High Priority (Next iteration)

3.5 Crop Rotation / Historical Data (Multi-Year Toggle)

Problem: A single year of CDL data is a snapshot. Field sales reps need to see 2-3 years of history to understand rotations (e.g., a field in grass this year may have grown broccoli last year).

Requirements:

- Ingest CDL data for **2022, 2023, and 2024** (minimum).
- Add a year toggle or slider in the UI (e.g., '24 | '23 | '22), and 'average'.
- At the parcel level, allow clicking a field and toggling between years to see what was planted.
- At the county level, show year-over-year acre change per crop.

Data source: USDA CDL archives (freely available for prior years in the same raster format).

3.6 Retailer / Distributor Location Pins

Problem: Field sales reps want to see ag-retail locations overlaid on the crop map so they can correlate retail presence with crop concentration.

Requirements:

- Support uploading or configuring a **point dataset** of retailer/distributor locations (CSV with name, lat, lon, and optional metadata like branch type).
- Display as map pins with a distinct icon layer.
- Allow toggling retailer pins on/off independently of crop layers.
- Clicking a pin shows the retailer name and any metadata.
- Support multiple location datasets (e.g., "Our Retailers," "Channel Partners," "Regional Offices") as separate toggleable layers.

Data source: GroNatural-provided CSV/spreadsheet. No public dataset; GroNatural will supply their own retailer and channel partner lists.

3.7 Improved Crop Groupings

Problem: Current crop categories don't align with how the sales team thinks about crops. Users want groupings based on how crops are grown.

Requirements:

- Restructure dataset to include categories such as:
  - Tree Fruit (apples, cherries, peaches, etc.)
  - Berry Crops (blueberries, strawberries, cane berries, etc.)
  - Vine Crops (grapes, etc.)
  - Annual Row Vegetables (broccoli, cauliflower, onions, lettuce, etc.)
  - Seed Crops (grass seed, etc.)
  - Nursery / Ornamental (Christmas trees, nurseries, etc.)
  - Field Crops (soybeans, corn, wheat — separate from specialty)
- Allow the user to expand/collapse groups in the sidebar (already supported in UI).
- Validate groupings with GroNatural before finalizing.

3.8 State-Relative Heat Map Option

Problem: National-level heat maps are useful for leadership but not for regional reps. A rep selling in the PNW wants to see where cherries are concentrated *within their region*, not relative to California's dominance nationally.

Requirements:

- Add a toggle for heat map reference frame: **"National"** vs. **"State"**.
- In "State" mode, the color scale normalizes to the selected state's total for that crop (i.e., what percentage of Washington's total cherry acres are in each county).
- In "National" mode, the color scale normalizes across all US counties (current behavior).

3.9 Monday.com (Exploratory)

Requirements:

- Allow exporting a filtered set of farms/parcels (e.g., "Top 10 cherry farms in PNW") to a CSV or directly to Monday.com
- Explore API integration with Monday.com
- At minimum, support a "Copy to Clipboard" or "Export CSV" action from any list view.
- Will be fleshed out in subsequent meeting with GroNatural VP of Operations

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P2 - Medium Priority (Subsequent iterations)

3.10 Multi-County / Area Selection with Summary

Requirements:

- Allow the user to click or lasso-select multiple counties.
- Display an aggregated summary panel: total acres by crop, number of farms, organic vs. conventional, across the selected area.
- Support saving named selections (e.g., "My Pacific NW Territory").

3.11 Rate-of-Change / Market Growth Metrics

Requirements:

- Once multi-year data is available (see 3.5), compute and display:
    - Year-over-year change in acres per crop per county (absolute and %).
    - A "growth trend" indicator (e.g., arrow up/down/flat) for each crop in each county.
- This supports the strategic question: *"Where is acreage increasing in our target crops?"*

3.12 Top Crops Ranking per County / State

Requirements:

- Display a "Top 5 Crops" or "Top 10 Counties" summary for any selected geography.
- Sortable by acres, by dollar value, or by growth rate.
- Accessible from the county detail panel and from a state-level summary view.

3.13 Soil Type Overlay

Requirements:

- Integrate **USDA Web Soil Survey** data.
- Display soil type as a toggleable map layer at the county or parcel level.
- Useful for agronomic context (e.g., understanding why certain crops are or aren't grown in an area).

Data source: USDA Web Soil Survey (free, publicly available).

P3 - Low Priority / Future Consideration

3.14 Sales Data Overlay

- Integrate internal sales data by salesman, county, state, and region.
- Requires GroNatural to provide sales data in a structured format.
- Display as an additional map layer or in the county detail panel.

3.15 Ag Chemical Label Lookup (Agrian Integration)

- One respondent requested chemical label lookup / Agrian integration.
- Explore feasibility; may be out of scope for the core mapping platform.
- Could be a link-out to Agrian from the crop detail panel.

3.16 Calibration Worksheet

- One respondent requested a calibration worksheet feature.

- Requires further scoping with that user to understand the use case.

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4. Data Accuracy Improvements

These are not features but pipeline/data-quality improvements required to address accuracy concerns raised in the survey.

Issue	Detail	Suggested Fix
PNW crop confusion	Satellite imagery struggles to distinguish blueberries, cane berries, grapes, Christmas trees, nurseries, and grass seed in the Pacific Northwest.	Document known CDL limitations per region. Consider supplementing CDL with NASS survey data for PNW permanent/seed crops. Flag low-confidence crops in the UI.
Missing cherries in WA	One respondent reported no cherries shown in Washington state despite significant production.	Verify CDL code mapping for cherries (CDL code 66). Cross-reference with NASS published acreage for Washington cherries. Fix if mapping error; document if CDL coverage gap. Preliminary probes (by national cherry % breakdown) have indicated data accuracy
Data vintage	Data is 2024; annual crops shift year to year.	Display the data year prominently in the UI. When multi-year data is available, default to most recent year.
Potential vs. planted	Users interpreted some data as "potential" rather than "actually planted."	Ensure all labels say "Planted Acres" and add a footer or tooltip: "Source: USDA Cropland Data Layer 2024 — satellite-classified planted acreage."

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5. UI / UX Improvements

Issue	Detail	Requirement
Variable naming	Some labels are unclear ("usable acres," density without context).	Audit all user-facing labels. Rename to plain-language terms. Add tooltips for any technical metric.

Issue	Detail	Requirement
Zoom detail	Users want more detail when zooming in to counties.	Ensure parcel-level data loads when zoomed to county level. Show individual field boundaries where data is available.
Data source attribution	Users want confidence in where data comes from.	Add a persistent footer or "About the Data" link citing USDA CDL, Organic Integrity Database, and any other sources.

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6. External Data Sources to Integrate

Source	Purpose	Priority	Notes
USDA Organic Integrity Database	Organic farm certifications and acres	P0	Partially integrated already
USDA NASS	Published crop acreage statistics	P1	Cross-validation and gap-filling for CDL
CDL 2022 & 2023	Historical crop rotation	P1	Same format as 2024; free from USDA
Client retailer/partner locations	Sales territory mapping	P1	GroNatural to provide CSV
USDA Web Soil Survey	Soil type context	P2	Free; large dataset
Client sales data	Revenue overlay	P3	Client to provide; format TBD
Client trial data	650+ trial locations	P2	Client has internal system
Agrian	Chemical label lookup	P3	Explore link-out only

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7. Success Metrics

Metric	Target	Measured By
Data accuracy confidence	Increase from 6.1 → 8.0+ (out of 10)	Follow-up survey after P0 delivery

Metric	Target	Measured By
UI comfort rating	Maintain 8.5+ (out of 10)	Follow-up survey
"Usable land" complaints	Zero	Follow-up survey; no mentions of misleading density
Organic data availability	Organic acres shown for all counties with certified farms	QA pass

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8. Out of Scope (This Phase)

- Mobile app or native application
- Real-time satellite imagery updates (we continue to use annual CDL releases)
- User authentication and role-based access
- Predictive modeling / ML-based crop forecasting (rate-of-change metrics are descriptive, not predictive)

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9. Appendix: Raw Survey Scores

Question	Avg Score	Range	n
Data accuracy confidence	6.1	3 – 9	11
Data robustness satisfaction	7.8	5 – 9	10
UI comfort	8.6	7 – 10	11